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Overcoming Vibration Limitations Through Integrated Infrasonic Technology in Industrial Sieving

Design and field validation of a modular infrasound retrofit platform for moisture affected screening

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Abstract

Sieving is a fundamental process in the construction, mining, pharmaceutical, and mineral sectors, however current practice of sieving is uneconomic and suffering from technical deficiencies such as screen blinding and blockage. Mechanical vibration is traditionally used by industry, however, it presents a thermal risk by friction for sensitive materials and may cause thermal damage.

This thesis examines the effect of infrasound assisted sieving, a non contact low frequency acoustic technology can improve on the separation efficiency where mechanical vibration fails on moisture affected material and whether it is technically and commercially feasible. Following the Double Diamond innovation process framework, the work begins by identifying various industry problems through qualitative interviews with leaders within different industries. The insights revealed that the main industry problem within traditional sieving was moisture and, as a result of the same, the time of operation for some industries (such as Nordic gravel production) was limited to some months in the year.

In response to the findings from research stage, the thesis development phase focuses on the design and validation of a modular retrofit platform. To validate the design experimentally a functional prototype has been tested at Skillingaryd field to analyze performance under high moisture environments. The trials have an estimated 50% throughput increase in moist moraine, with stable operation at an infrasonic frequency of 16.60Hz with peak pressures at around 17 kpa. A two part Design Failure mode and effects analysis (DFMEA) was completed in advance of field trial and based on lessons learnt during the trial, and found the fail safe system (RPN 504) and control electronic dust ingress (RPN 432) as the key risks.

The modular platform developed provides an easy and scalable integration between existing machinery and infrasonic modules. This enables significant reductions in energy consumption and significant extensions of the usable industrial sieving period during the year. In addition, the total business case with market size, business economic models, go to market plans, etc., shows an addressable market in the Nordic countries of SEK 2.8 billion with the target to payback the machines within 3-4 years for a customer.

Keywords

Infrasound, industrial sieving, modular design, retrofit strategy, DFMEA, business case, Double Diamond, acoustic fluidisation.

Sammanfattning

Siktning är en grundläggande process inom bygg-, gruv-, läkemedels- och mineralsektorerna, men nuvarande siktmetoder är ekonomiskt ineffektiva och lider av tekniska brister som igensättning och blockering av siktdukar. Mekanisk vibration används traditionellt inom industrin, men kan medföra termiska risker genom friktion för känsliga material och orsaka värmeskador.

Detta examensarbete undersöker effekten av infraljudsassisterad siktning, en kontaktfri, lågfrekvent akustisk teknik, på separationseffektivitet samt metodens kommersiella och tekniska genomförbarhet. Med hjälp av Double Diamond-innovationsramverket inleds projektet med att identifiera industriproblem genom kvalitativa intervjuer med ledare inom olika sektorer. Insikterna visade att det primära problemet inom traditionell siktning var fukt. Till följd av detta var driftsäsongen för vissa industrier (exempelvis nordisk grustillverkning) begränsad till endast några månader per år.

Som svar på undersökandefasen fokuserar avhandlingens utvecklingsfas på design och validering av en modulär retrofitplattform. En funktionell prototyp fälttestades på Skillingaryd för att analysera prestanda under hög fuktighet. Försöken producerade en 50-procentig ökning av genomströmningen för fuktig morän, med stabil drift vid en infraljudsfrekvens på 16,60 Hz och trycktoppar på 16–17 kPa. En Design-FMEA genomfördes i två steg, en inledande konceptanalysfas och en reviderad analys baserad på fältobservationer. Under analysarbetet identifierades säkerhetsavstängning (RPN 504) och damminträngning i styrelektroniken (RPN 432) som de högst prioriterade riskerna.

Den utvecklade modulära plattformen erbjuder en skalbar lösning som förenar befintlig utrustning med effektiv infraljudsteknik för att minska energianvändningen och väsentligt förlänga driftsäsongen för industriell siktning. En fullständig affärsanalys indikerar en adresserbar marknad på cirka 2,8 miljarder SEK i Norden enbart, med en målsatt återbetalningstid för kunden på 3–4 år.

Nyckelord

Infraljud, industriell siktning, modulär design, retrofitstrategi, DFMEA, affärsmöjlighet, Double Diamond, akustisk fluidisering

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Lastly, I wish to thank KTH for providing an excellent learning environment and my colleagues for support and valuable feedback through seminars.

List of abbreviations

The following key terms and abbreviations are used throughout this thesis:

Term / Abbreviation	Definition
Infrasound	Sound waves with frequencies below 20 Hz, inaudible to the human ear but capable of significant particle displacement.
Acoustic Fluidisation	The phenomenon where a bed of cohesive particles behaves like a fluid under the influence of acoustic energy.
Blinding / Clogging	Obstruction of sieve apertures by particles, leading to reduced throughput and separation efficiency.
Van der Waals Forces	Short-range intermolecular attractive forces that cause fine particles to adhere to one another and to sieve mesh.
Capillary Liquid Bridges	Water films that form between particles in moist conditions, creating strong cohesive bonds that resist sieving.
Double Diamond	A design framework by the UK Design Council consisting of four phases: Discover, Define, Develop, Deliver.
DFMEA	Design Failure Mode and Effects Analysis — a systematic method for identifying potential failure points in a design.
RPN	Risk Priority Number — the product of Severity, Occurrence, and Detection ratings (each 1–10) in an FMEA.
HaaS	Hardware as a Service — a revenue model where equipment is leased rather than sold outright.
CAPEX	Capital Expenditure — upfront investment in physical assets.
Pulsator	The infrasound-generating device that creates rhythmic pressure pulses, core to the Infrasonik AB system.
EMS	Electronic Management System — used here to regulate blower RPM during field testing.
MoSCoW	Prioritisation method: Must Have, Should Have, Could Have, Won't Have.
API	Active Pharmaceutical Ingredient — the biologically active component of a drug.
SAM	Serviceable Addressable Market — the portion of the total market reachable by the company's product.
TAM	Total Addressable Market — the overall revenue opportunity if full market share were captured.
SOM	Serviceable Obtainable Market — the realistic share a company can capture in the near term.
IP6X	IEC enclosure rating indicating complete protection against dust ingress.
VSMA	Vibrating Screen Manufacturers Association — publishes standard screen capacity calculation methods.
kPa	Kilopascal — unit of pressure (1 kPa = 1000 Pa); used here for acoustic pressure amplitudes.
FMEA	Failure Mode and Effects Analysis — the general method; DFMEA denotes its application to a design.
RPM	Revolutions per minute — used here for the blower speed set by the EMS.
ROI	Return on investment.
t/hr	Tonnes per hour — unit of screening throughput, measured as mass flow at the discharge point.
Hz	Hertz — cycles per second; infrasound is defined as below 20 Hz.

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1. Introduction

In many of the manufacturing industries across the world the process of sieving and separating industrial materials is central such as mining, food processing, Pharmaceutical, and many more. However, the technology used to accomplish this task has not drastically changed over many years [1].

Production sites typically use a large mechanical vibrating screen to carry out the sieving of the materials this technology, however, is easily affected by moist and sticky materials when the screens block up and stopping production until they are dried [2].

The purpose of this thesis is to demonstrate how the low frequency infrasound application was applied in practice for industrial sieving as a modification of the Infrasonik AB existing core technology rather than an evolution, because the core product remained practically unchanged whilst the new mode of sieving technique was developed. The application was considered to have the potential for high value alternative because it could address limitations which are commonly associated with conventional sieving process and this should alleviate the problems associated with humidity such as blinding as well as the high Mechanical wear which means a fast production, lower cost and higher quality guarantee. The thesis investigates the practical implementation of this application and evaluates its potential as an alternative approach to industrial material sieving or seperation.

1.1 Background & Problem

This work derives from an inquiry from a customer to Infrasonik AB. Infrasonik AB knowledge is centered in infrasound applications to drying, cooling and cleaning of soot. Infrasonik took the inquiry literally and built and constructed a small scale model of the sieve using the current Infrasonik AB knowledge and applied the method on the particles delivered by the customer. It remained uncertain if this would work at larger scale, which can be covered in this work. Using the acquired experience from the preliminary testing Infrasonik design first sieving machine. We tested the method in Skillingaryd which is now referred to as InfraSieving method.

Industrial sieving is generally based on mechanical vibration. More recently, mechanical vibration is achieved through high frequency sieve vibration system,

otherwise known as ultrasonic sieving, where piezoelectric actuators vibrate a sieve mesh at a frequency greater than 20,000Hz [3].

These methods, in conjunction with mechanically vibrated sieve, are both highly problematic in dealing with wet, sticky or cohesive fine powders. The causes of the latter being the formation of capillary liquid bridges at the presence of moisture, which binds particles to each other in such way that they block sieve apertures called mesh blinding [1], [2]. It has been reported that a large fraction of organisations within regulated industries encountered line clearance failures through the binding of particles, creating risks, both economic and in terms of safety, to their production process [4].



Figure 1.0 – Blinding (Infrasonik AB source)

Infrasonik AB found the technology gap that is the inability of current technologies to efficiently sieve ultra fine and/or wet materials. With mechanical vibration, the displacement is not enough to free the wet particles from the moisture bonds without damaging the mesh [1], [2]. With high-frequency sieve vibration systems the mesh surface produces heat through friction which damages temperature sensitive ingredients, such as pharmaceutical actives [3]. Low frequency Infrasound has been proposed as potential solution to these limitations.

1.2 Purpose & Objective

This research will attempt to demonstrate how infrasonic sieving can address both the limitations of mechanical sieving and those of high-frequency sieve vibration and determine its viability for technical and commercial implementation in industry. Infrasound however is below 20 Hz and causes macroscopic particle

displacement (see Eq. 1, Section 2.2), as a completely different principle of operation from high frequency vibration methods [5].

This thesis will investigate the feasibility of infrasonic sieving as high-performance alternative. Mechanical effects on sieving due to infrasound are analysed. A modular design for addressing generic industrial problems is proposed.

The project intends to:

- Research market demand and segmentation and to determine an effective market positioning for infrasonic technology.
- Design a modular device suitable for seamless integration into current manufacturing lines of the construction, mining, food and pharmaceutical industries.
- Technically validate the InfraSieving method through laboratory and field experiments.
- Conduct a two-stage DFMEA — before and after the Skillingaryd field trial to identify and mitigate design risks
- Formulate a quantified business case and market entry strategy.

1.3 Delimitations

Due to time and resource limitations, the thesis is limited in scope and aspects of this scope include the difficulty in sourcing sufficient interviewees with good practical experience in screening and limitations to the collection of a statistically relevant body of experimental data in the project's timeframe. The scope in terms of industry and geography was reduced to a single due to timeframe, particular industry in Sweden, that of mining/gravel. Applications within food or pharmaceuticals are considered an area for future work.

2. Theoretical Background

The principles needed to understand the scope of the project are laid out within this chapter, namely, industry demands, the technological deficit in sieving and the physics of acoustic fluidisation.

2.1 Introduction to Industrial Screening

Screening, also known as sieving, is a common and essential operation in mining, agriculture, medicine, food processing, building construction and various other industries [2]. The aim of screening is to increase the economical value of a crushed or granular substance by separating it into different size grades. Industrial screening processes suffer from a conflict between efficiency (correctness of separation) and capacity (speed of the processing) [2].

Research on linear vibrating screens has shown that screen efficiency increases with length, but that diminishing returns are reached quickly beyond which the increase in cost would outweigh the efficiency increase. As a result, new techniques with high capacity without clogging of screens or reduced accuracy of separation are wanted [1].

2.2 Sound, Infrasound, and Ultrasound

Infrasound is sound with a frequency less than 20Hz [5]. There are two quite different physical characteristics that contribute to the effectiveness of infrasound in industrial fluidisation, which should not be confused.

The first is low loss in propagation [5]. The wavelength and frequency of a sound wave are inversely related (wavelength = speed of sound / frequency), thus infrasound consists of waves with extremely long wavelengths, of about 34 m for 10 Hz and 343 m for 1 Hz. Energy of this sort can be readily transmitted through air and gas systems without excessive attenuation and thus consistently supplied over the whole length of the sieve deck [5].

The second (and more relevant for sieving) characteristic is that there is large amplitude of particle displacement. In an acoustic wave the maximum particle displacement is at the pressure node (the quarter-wave point). As the amplitude is inversely proportional to the frequency, then at low frequencies the amplitude is on a macroscopic level, for example ranging from centimetres to just under a meter in industrial infrasound systems. This is the fundamental reason why infrasound

can fluidise materials beds. The relationship between acoustic pressure amplitude P and particle velocity amplitude u is:

u = P / (ρ · c) (Eq. 1) (fix particle displacement)

Where ***ρ*** is air density (kg/m3) and ***c*** is speed of sound (m/s). For a given pressure amplitude, lower frequencies yield higher particle velocity, and displacement lower is the frequency given the particle pressure amplitude. This is why infrasound has macroscopic motion of material that high frequency system does not [5].

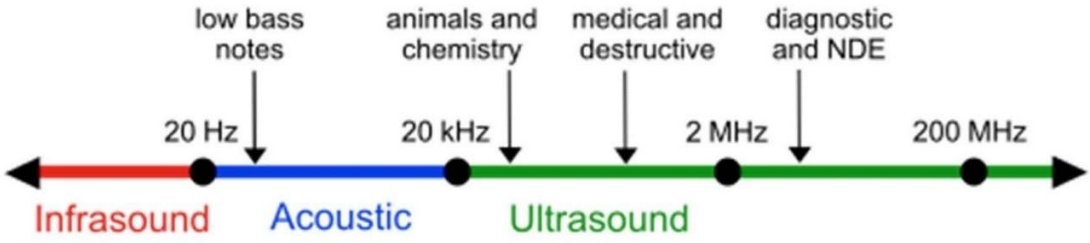


Figure 2 — Infrasound Representation

High frequency sieve vibration systems are commercial systems referred to as ultrasonic sieving [4]. The principle of operation is a piezoelectric actuator mounted onto the frame of the sieve mesh; this mechanism provides mechanical vibration of the sieve to in excess of 20,000 Hz. The systems do not employ acoustic waves traveling through the air but mechanical vibration of the physical sieve itself [3].

Very high frequency causes very low particle displacement, in the micrometer range. While low displacement may be sufficient to fracture adhesive forces binding fine dry powders it is inadequate for the movement of dense, wet or moisture laden beds of material. The heat produced by direct friction of rapid mechanical oscillation of the mesh can degrade some sensitive compounds such as pharmaceutical active ingredients [3].

There is a marked difference to infrasound assisted sieving both in terms of mechanism (mechanical oscillation of the mesh versus acoustic waves traveling through the air) and scale of displacement (micrometers versus centimeters to meters) and they are not mutually interchangeable.

2.3 Industry Needs vs Technology Gap

The mechanical vibratory sieving market also presents a substantial gap between capabilities and needs. High purity, fine separations are essential to the mining, food, and pharmaceutical industries, while traditional mechanical vibration causes mesh blinding (mining) and thermal degradation (pharmaceutical). This gap represents the market for a non-contact, low frequency separator [2].

Table 1 — Traditional requirements mechanical sieving compared to modern industrial requirements

Industry	Current Need	Current Technology	Why Traditional Fails	Evidence
Mining (LKAB, Boliden)	High-volume throughput of wet, sticky magnetite and clay.	High-force mechanical vibrating screens	Blinding: wet fine material sticks to the screen, blocking holes and requiring frequent downtime.	Metso [28]; Wills & Finch [27]
Food (Nestlé, Arla)	100% purity and high-speed flow of fatty/moisture-sensitive powders.	Vibratory circular separators	Agglomeration: mechanical shaking packs fine powders together. Metal-to-metal friction creates contamination risk.	Russell Finex [29]
Pharma (AstraZeneca)	Zero thermal degradation of heat-sensitive Active Ingredients (APIs).	Mechanical shaking sifters	Thermal friction: rapid vibration generates heat that can alter the chemical structure of sensitive drugs.	Telsonic [30]
Chemicals (BASF, Dow)	Precision sizing of ultra-fine polymers without dust suspension.	Tumbler screens (without acoustic assistance)	Material float: fine particles remain airborne during violent shaking rather than passing through the mesh.	Allgaier Process Technology [31]; Wills & Finch [27]

The systematic narrowing of this broad, multi-industry problem down to the specific Swedish mining and gravel scope tested in this thesis is presented as a funnel delimitation diagram in Section 4.1, together with the methodological rationale for using this framework.

2.4 Acoustic Fluidisation Theory

An important drawback that fluidisation technology has faced is achieving the transition of particles from a static particulate bed to a fully fluidised state. Gas-solid fluidisation works well with the large sized particles but struggles with fine and ultrafine powders as the interparticle cohesive forces that exceed the gravitational forces lead to clogging, Blinding and slugging which can cause fluidisation to be highly inefficient [7].

Therefore, the concept of acoustic fluidisation as a niche treatment, as defined by [7], where high intensity sonic waves in the infrasonic or low-frequency sonic region are used to break cohesive bonds and cause even bed expansion. Bubble free fluidisation of cohesive powders in an acoustic field is the terminology given by [7].

2.4.1 Dominant Forces in Cohesive Powders

The primary cause of agglomeration in fine powders is the van der Waals force. For two spherical particles of identical radius r separated by a distance d, the attractive force is:

Table 2 — Dominant interparticle forces in fine cohesive powders

Force Type	Mechanism	Typical Magnitude (N)	Sensitivity to Environment
Van der Waals	Fluctuating dipole–dipole interactions	10^{-12} to 10^{-7}	Low sensitivity to moisture/temperature
Electrostatic	Surface charge transfer and Coulombic attraction	10^{-12} to 10^{-6}	High sensitivity to humidity
Capillary	Liquid bridge formation at contact points	10^{-8} to 10^{-5}	Extremely sensitive to humidity
Gravitational	Mass-dependent attraction	10^{-13} to 10^{-6}	Independent of environment

$$F_{vdw} = (Ah \cdot r) / (12 \cdot d^2) \qquad \textbf{(Eq. 2)}$$

Where A_H is the Hamaker constant. As particle radius decreases, the ratio of this attractive force to the particle's gravitational weight increases exponentially, leading to the formation of stable agglomerates.

The data shows that for particles those with a mean diameter of less than 30 micrometres the cohesive forces can be three to four orders of magnitude stronger than the gravitational force [8].

This imbalance shows that gas flowing through such a bed cannot easily separate individual particles instead it lifts the bed as a single plug or carves channels through the material [9].

2.4.2 The Role of Oscillating Air vs. Oscillating Machines

In acoustic fluidisation the fluidising medium that is air is subjected to acoustic oscillations, while the material bed and its container remain mechanically stationary. Unlike standard vibratory fluidised beds which rely on a mechanical actuator which shakes the container, transmitting energy through its walls to the particles, it is difficult to transmit large amounts of energy into the particles with this type of setup, particularly for high columns where vibrations will be transmitted weakly at the top levels [10].

Acoustic fluidisation relies on the transmission of sound waves through the gas between particles and provides an input of energy to the force chains that link them together, this is a more holistic and less invasive energy input mechanism compared to bulk mechanical agitation [7]. This mechanism can reduce wear and fatigue in the system significantly due to the lack of bulk motion.

2.4.3 Mechanism of Infrasonic and Low-Frequency Intervention

Acoustic fluidisation depends upon the frequency and amplitude of the sound waves. Studies done on nanoparticle agglomerates at NJIT experimented with low-frequency sonic waves of frequencies between 20-600 Hz and found that results steadily improve within this frequency range as frequency decreases [8].

No frequencies below 20 Hz (infrasound) were tested, but the trend of increasing success as frequency decreases suggests that infrasound will achieve greater fluidisation than what is displayed. The use of low frequencies translates to long wavelengths and large particle displacements, allowing for the macroscopic force needed to simultaneously overcome capillary liquid bridges and van der Waals forces.

3. Literature Study

This chapter shows the results of the literature study of core screening difficulties and solution options, infrasound innovation, and product development. Chapter 2 discusses conceptual context and definitions, this chapter discusses what has been found in research literature.

3.1 Core Challenges: Blinding, Moisture, and Agglomeration

The main points of research findings on the three major reasons of sieving failure are summarized in the following sub-sections:

3.1.1 Blinding

Siliveru and Ambrose Siliveru and Ambrose [11] produced a comprehensive predictive model for particle separation and sieve blinding during wheat flour sifting and conclude that "blinding is inherent in mesh-based separation." Their model confirms that blinding represents a fundamental limitation of aperture-based separation methods, not merely an operational inconvenience [11].

3.1.2 Moisture

Moisture is another major factor causing blockages. It is known that in studies on sand screening machines, the sieving time can be greatly affected by moisture content and the efficiency can be reduced by 15% [1].

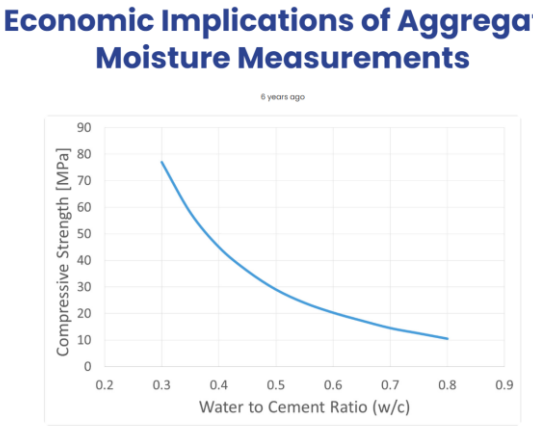


Figure 3 — Moisture Content

For agricultural studies soil moisture is critical factor in controlling particle fragmentation and thus dry sieving is an unreliable procedure unless soil sample is dried prior to this operation. Moisture contents and wet aggregates affect concrete strength badly and increase fuel consumption in drying [1], [12]

3.1.3 Agglomeration

Studies using ultrasonic-assisted screening show that fine powders clump together due to electrostatic and van der Waals attraction and that small powders are treated as large particles and will not pass through the mesh. This is one of the major causes of failure in the separation of fine, dry powders [3].

3.2 Current Solutions

Two most common technological systems in industrial screening are linear mechanical screens and high frequency vibration systems. Key results drawn from the research literature are summarized below:

3.2.1 Mechanical Linear Screens

Research shows it has been found that Mechanical Screen efficiency is most dependant on setting the vibration frequency and screen incline angle appropriately, and for gravel applications the best frequency and angle were approximately 7 Hz and 15 degrees respectively [2].

It has also been shown that mechanical vibration will not effectively break wet and sticky aggregates [3].

3.2.2 Ultrasonic Technology

In addition to Chen and Yao's findings on the ultrasonic-assisted screening of fine powder material, Leventhall [3] demonstrated that high-frequency vibration of the mesh removes adhesion effects in dry fine powders [12]

The same study also states that although this method effectively screens dry fine powders it cannot be implemented in moist, heavy, or dense bed situations due to very small mesh displacements (micrometer range).

3.3 Infrasound as an Innovation

These last three subsections are summaries of what has been documented in the research study concerning infrasound's direct means of achieving better sieving.

3.3.1 Infrasound Physics: High Displacement

According to work by [13] involving the infrasound cooling of a steel plate at KTH, maximum particle velocity amplitude, gained via a low frequency and long stroke length results in maximum physical transfer effects [5], [13].

This is relevant to the sieving application, as it means particles and air are moved across cms, not microns, thus gaining the physical force to move wet aggregates.

3.3.2 Fluidisation Effect

Natarajan Natarajan [14] experimentally optimised an infrasound dryer used for wheat, showing that a static solid bed of grain became a dynamic fluid-like state of grain by acoustic fluidisation [14]. This is experimental, not theoretical. The same fluidisation mechanism is expected to be present for cohesive grains on a sieve deck.

3.3.3 Boundary Layer Disruption

Both biochar production research and research on wheat drying have verified that infrasound produces turbulence in which the boundary layers-still air layer and moisture layer-on the particles are destroyed [1], [4].

When applied to sieving, such turbulence not only favors the drying of particles from the surfaces but also favours the particles sticking to themselves and to the mesh.

3.4 Product Development and Implementation Strategy

The following sub-sections are devoted to summarising results pertinent to industrial, commercial and technical exploitation of infrasound technologies

3.4.1 Innovation Management

There is also the difficulty of integrating infrasound in industrial equipment. Conventional linear development cycles are inadequate. The development of sustainable iterative circles, that means fast prototyping and testing cycles, should help industrial systems evolve according to the needs of customers [15], [16].

It has also been shown in the manufacturing industry that external partners such as suppliers, university, customer can improve innovation by open innovation [17].

3.4.2 Market Strategy

Market segmentation and product differentiation research is in favour of targeting infrasound-assisted sieving to market segments in which current solutions are lacking rather than engaging in price competition in segments where mechanical vibration is performing adequately [21], [22].Comparison based on user-generated content and sentiment analysis could be performed to identify inadequacies in the performance of existing mechanical vibrating screens for the B2B market [23].

3.5 Research gaps

The literature summarized in this chapter, identifies four specific gaps that is addressed by this thesis, which provide the basis for the research questions already presented in section 3.5. Instead of restating these questions in this chapter, it will instead highlight the gaps the research questions are seeking to fill with regards to the existing literature.

Gap 1 (underpinning RQ1): Section 3.3 highlights that better fluidisation is achieved with decreasing sonic frequency to the infrasonic domain however, no previous publication specifically investigates infrasound (below 20Hz) on an actual sieve screen under high-moisture conditions; this thesis presents the lacking empirical link.

Gap 2 (underpinning RQ2): No existing study, using Infrasonik AB's previous applications of infrasound technology (drying, cooling and soot cleaning) and the related modular/open-innovation literature (section 3.4.1), explains the packaging requirements for an infrasound-generating module as a retrofit to standard sieve machinery; there are no existing product design requirements literature for this combination.

Gap 3 (underpinning RQ3): Shows a decrease in energy input and in clogging for both dry, fine powders using ultrasonic assisted sieving (section 3.2.2). There is an absence of quantitative data concerning the benefits for 2-4mm particles in high moisture conditions using infrasound.

Gap 4 (underpinning RQ4): While FMEA and market-strategy literature (sections 3.4.2, 3.4.3) do outline applications for existing industrial machinery, no similar combined technical-risk and go-to-market framework has been presented specifically for this infrasound retrofit product.

These four gaps directly lead to the four research questions presented in section 3.5. These were presented in section 3.5 and answered in section 8 (Conclusion).

3.6 Research Questions

The following research questions guide this thesis and will be answered in the Conclusion (Section 8).

- RQ1: How can infrasound overcome environmental and moisture related problems that currently inhibit traditional sieving methods in industry?
- RQ2: What specific design requirements must be met to enable modular retrofitting of infrasound technology into existing production lines?
- RQ3: To what extent can infrasound induced fluidisation reduce energy consumption and prevent mesh clogging when sieving 2–4 mm particles in high-moisture conditions?
- RQ4: What are the core requirements for a go to market strategy that bridges the gap between infrasound capability and the operational needs of moisture affected industries?

4. Methodology

This chapter describes the research approach adopted in the thesis, the workflow of the project, the design methodology, data acquisition, the qualitative and experimental setup that were used for both lab testing and field testing.

4.1 Research approach

The thesis addresses a question that is simultaneously technical and commercial: whether infrasound-assisted sieving works under adverse industrial conditions, and whether it can be brought to market as a retrofit product. Neither half can be answered by a single method. Establishing whether the technology performs requires controlled measurement of throughput, frequency and acoustic pressure; establishing whether the performance matters require access to the operational reality of the people who run screening plants, which is not documented in the published literature.

The thesis addresses a question that is both technical and commercial whether infrasound assisted sieving works under harsh industrial conditions, and whether it can be brought to market as a retrofit product. This can not be answered by a single method. Establishing whether the technology performs require controlled measurement of throughput, frequency and acoustic pressure , establishing whether the performance matters require access to the operational reality of the people who run screening plants, which is not documented in the published literature.

There is need of mixed method design and therefore adopted combing a qualitative strand and a quantitative experimental strand. The qualitative strand includes semi structured interviews conducted in the discover phase and analysed thematically, to complemented by an internal strategy survey at Infrasonik AB during the delivery phase. The quantitative strand includes of lab sieving trials, instrumented pressure management, and a two day field trial at an operating site. A structured risk assessment is also built by translating interview derived requirements and field observed behaviour which were translated by strands linked to a two stage Design Failure Mode and Effect Analysis (DFMEA).

These two threads of research are serial and exploratory, not simultaneous: First identified the problem and derived the specification from a process of qualitatively describing current needs and second investigated whether the solution met the specifications experimentally. This is necessary if to follow the logic of the Double Diamond developed in Section 4.2 where understanding the problem is,

conceptually (and strategically), treated distinctly from solving the problem. Qualitative is not there to yield statistically robust claims about screening providers, but to prove that the problem of concern is real and perceived by industry personnel to require attention and that the requirements Specification is based upon practical experience, not designer imagination.

4.2 The Double Diamond Process

This thesis adopts the Double Diamond design framework (Design Council, UK, 2005) as a structured approach to guide the product development process. Technological feasibility and market requirements are subsequently evaluated through the technical and market validation activities presented in this thesis. The Double Diamond includes four stages of design: discover (finding out about the problem space using interviews and observation), define (synthesising information and producing problem statements and requirements), develop (prototyping and iterating on solutions) and deliver (creating and refining the solution). It was chosen because it separates the process of understanding the problem from the process of developing the solution; preventing a tendency in technology-push projects to create solutions for non-problems.

To make the scoping process itself transparent and traceable, a funnel delimitation diagram is used below to show how the broad, multi-industry problem introduced in Chapter 1 and Section 2.3 was progressively narrowed to the precise research scope investigated in this thesis.

Details about each stage and what methodologies that are applied in that stage are presented as below - for the Discover-stage, qualitative methods in Section 4.3 and experimental interventions in Develop- and Deliver-stages in Section 4.4 and risk analysis across stages Define and Deliver in Section 4.5.

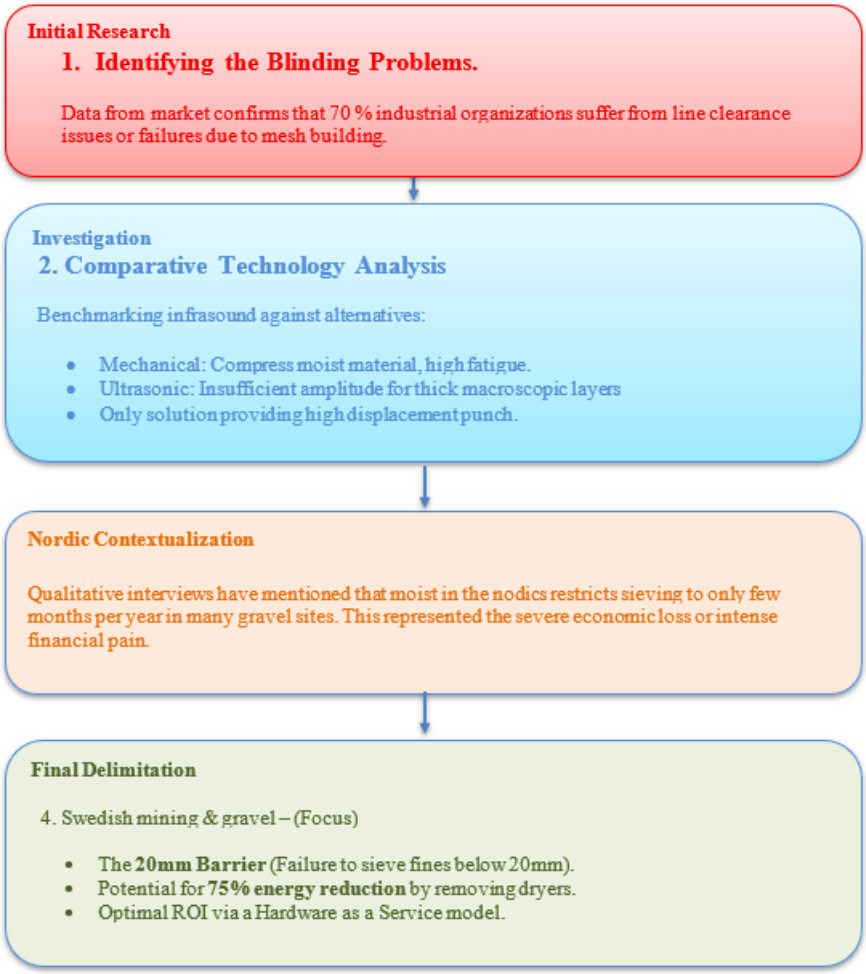


Figure 4 — Strategic Delimitation flowchart

Figure 4 is an illustration of the funnel delimitation diagram; a diagram that formalises a systematic way of showing how the broad research scope has been progressively constrained to a narrower research problem through stages of definition. The funnel format is proposed in system engineering and design research methods to provide the reader with a transparent and traceable path from the initial broad industrial problem, down to the precise and justifiable research scope [6]. The funnel format was selected for three reasons.

Firstly, the problem was initially framed on a wide-ranging, multiple industry issue (mesh blinding in mining, food, pharma, and construction) and needed a way to be broken down systematically.

Secondly, the four stages of Initial Research, Investigation, Nordic Contextualisation, and Final Delimitation represent the steps of the actual decisions taken in the Discover section of the double diamond framework and thus presents an honest way to show how the scope was set.

Thirdly, the funnel gives a better understanding than prose alone how the logic of delimitation, towards narrowing the scope to solely Swedish mining and gravel as the testing ground despite a much larger market, works.

Stage 1 – discover (month 1): Identifying industry pain points (screen blinding, material clumping, dust leak, operational downtime costs etc.) through semi-structured interviews with potential Infrasonik customers (plant managers, production managers, process engineers etc.). Establishing competitive Infrasonik vs Ultrasonic / mechanical vibration technology benchmark analysis.

Stage 2 – define (month 2): Identifying market segments (based on severity of problem, addressability etc). Designing new build & retrofit general system architecture and components. Preparing preliminary "pre-build" DFMEA to explore potential conceptual risks.

Stage 3 – develop (month 3): Designing standalone portable infrasound module. CAD designing with Infrasonik in SolidWorks / Fusion360. Infrasonik AB Team co design workshops.

Stage 4 – deliver (month 4-5): On-site field testing Skillingaryd (April 2026). Developing post-field trial DFMEA based on observations. Creating a business case including ROIs & contrasting CAPEX / HaaS revenue streams. Generating final BoM & producing assembly instructions & retrofit guide.4

4.3 Qualitative study

The qualitative aspect consists of two different strands with their own objectives and their own weight of evidence. The discover process created an internal set of informal in-depth interviews to explore industrial problem. The commercial recommendation developed during the discovery phase was tested within an internal strategy survey in the deliver phase in order to determine how it aligns with current corporate strategy. I separate these from each other since they involve a distinct range of respondents with respectively different point in the lifecycle of my research so with different evidence quality.

4.3.1 Interview Design

Semi-structured interviews rather than structured questionnaire Since the purpose of the Discover stage was exploration, it was judged it is best to avoid structured

interviews. The specific pain points related to industrial screening were not sufficiently well understood to define fixed response items, and a structured instrument may risk reinforcing the technology supplier's prior beliefs rather than the primary concerns of practitioners. My thought that Semi structured interviewing has value is nicely illustrated by the Seasonality finding discussed in Section 5.2; The need to screen dry seasons became apparent only after an unprompted reply was obtained, but I was only prompted when the question appeared on my list, not at any prior juncture.

There have been three tailored versions used to interview the key stakeholders rather than a single generic one, built around a similar core topic structure – allowing interview question tailored to each participant's language.

These three versions are summarised in Table 3.

Table 3 — Interview guide versions and topic coverage

Instrument version	Directed at	Topic areas covered
Guide A – end user	Operators of mechanical or ultrasonic screening	Current setup and material behaviour; blinding and cleaning downtime; mesh breakage and consumable cost; moisture, ice and seasonal effects; heat sensitivity of product; retrofit feasibility; energy use; simple ROI
Guide B – technology developer	Founder and technical staff, Infrasonik AB	Origin and intended problem; comparison with mechanical and ultrasonic alternatives; effect on mesh life; retrofit versus new build; workplace noise; agglomeration behaviour; cold and moisture performance; energy and cleaning-time savings
Guide C – metal powder producer	Process engineering, Sandvik Coromant	Existing screening technology; difficult powders; process bottlenecks; mesh fatigue; blinding frequency and cleaning time; consumable lead time and cost; moisture and agglomeration; thermal sensitivity of powder
Guide D – Steel manufacturing company	SSAB	Seasonal problems, moist material, and blinding.

They shared a common core of topic questions namely the 'screening process in use' and associated' material' performance characteristics; the 'screening failure mode'; 'equipment downtime'; and any' consumables costs associated'. Further, the impact of and concern 'Seasonal' variation; 'moisture'; and 'temperature'; and then third the 'conditional requirements that any new technology must fulfil'. Both of their question wording is open and of no leading nature.

It was found that, the most useful approach whilst interviewing any end user was to let them discuss what their current issues were prior to any questioning regarding infrasound for the proposed screen technology

4.3.2 Participant selection

Selection of Participants Participants were selected using a non-random, purposeful approach, due to the small size and high degree of specialisation of the population and because information richness rather than statistical representativeness was the primary sampling goal. The three criteria for selection were: - The organisation involved conducted industrial screening on a routine rather than an occasional or a laboratory basis, The individual approached had direct, day-to-day insight into the performance and availability of the screening operation, and - The site of the organisation's activities fell within the Nordic circumstances defining the limits of this research. The first criterion had one intentional exception in the case of the founder of Infrasonik AB, for whom this research role of technological informant rather than screening end-user was requested, an objective was to compile the design rationale and genesis of the infra-sound generation system that had not been otherwise documented. Candidates were found from the commercial contacts of Infrasonik AB and by directly contacting companies involved in Swedish gravel, mining and minerals extraction. Eight interviews were pursued and four successfully secured and this deficit and the likely impacts are detailed in section 4.6 in the analysis section. Selected participants are summarised in Table 4.

Table 4 — Interview participants

Ref.	Role and organisation	Sector / relevance	Format
P1 – “Per-Erik”	Owner and operations lead, Skillingskog AB	Gravel and stone material production for road building and concrete foundations; end user of mechanical screening	Semi-structured recorded and transcribed with verbal consent
P2 – “Mats”	Founder, Infrasonik AB	Technology developer; 45 years of infrasound generator development. Internal informant — see Section 4.6	Semi-structured recorded and transcribed with verbal consent
P3 – “Arvind Kartha”	Senior Engineer, Sandvik Technology	Mechanical Engineering	Semi-structured recorded and transcribed with verbal consent
P4 – “Per”	SSAB - Manager	Production and operations	Semi-structured recorded and transcribed with verbal consent

4.3.3 Thematic analysis

Interview data were subjected to reflexive thematic analysis [33], applying the six-phase procedure described by Braun and Clarke [32]. The method was selected because it is suited to identifying patterned meaning across a small number of rich accounts and does not presuppose the sample size required by content-analytic or statistical approaches. The steps involved in applying the methodology were as follows: all transcripts were read through in their entirety in order to establish familiarity with the data set. Next, initial codes were generated inductively by providing labels to each section of transcript that represented a description of the screening problem, a consequence thereof or a criterion for a satisfactory solution (codes were generated in the data rather than from pre-existing theory). Candidate themes were formed by collating inductively derived codes based on the underlying concepts that they represented; these candidate themes were refined by revisiting the coded extracts in order to both check internal homogeneity and mutual exclusivity of themes and were subsequently named, defined and illustrated with clear excerpts.

Two themes resulted from this process; these are entitled The Pain-Points of Industrial Screening and Technical Differentiation, consisting respectively of four and three codes relating to respectively the circumstances in which conventional screening methods fail and the characteristics of infrasound relative to those of mechanical and ultrasonic methods. These themes, consisting of codes and illustrative examples are presented in sections 5.2 and 5.3.

The analysis was carried out by a single researcher utilizing a manually structured coding table, thereby preventing any possibility of testing inter-rater reliability. To a degree this deficiency was overcome by discussing the definitions applied to the codes and themes with the supervisor representing the industrial sector Infrasonik AB, and by including for each reported code a verbatim quotation of the relevant section so that the reader may examine for themselves the inferences made from data to theme; the weaknesses of a single-coders analysis are discussed in section 4.6.

4.4 Design & Experimental Setup

The requirements specification for the prototype was derived from two sources: From the literature study (Chapter 3) for the identification of the principles by which screen mesh blocking works, and the qualitative interviews described (4.3) which investigated how that blocking happens in a production environment, and what a working replacement would need to do, three functional requirements emerged: Energy would need to be coupled to the bed of material without physical interaction. The use of vibrations, as performed by existing vibratory screens was shown to be a significant factor in mesh wear and bearing longevity in the interviews. There needed to be sufficient displacement to allow the breakdown of capillary bridges which form an otherwise significant part of cohesion and bed structure above some characteristic moisture content (above ~ 5-10% as per 2.4), but not simply enough agitation to merely stir the bed.

There needed to be sufficient versatility to be retrospectively fitted to currently installed screening plant: only one interviewee seriously contemplated replacing his entire screening facility as commercially viable.

The research described then tests whether a prototype built to this set of functional requirements satisfies them. Each section of the test work serves a distinct research objective: Section 4.4.3 examines the results of laboratory experiments to test RQ3. They determine by measuring the screen throughput on reference beds, how the introduction of ultrasonic assistance alters it compared to the baseline (no assistance). Section4.4.4 examines the results of the full scale field test to test RQ1.

They determine the throughput under real production conditions, through moisture affected beds.

Underlying these experiments (4.4.2, to demonstrate this) were measurements to prove that the output signal was stably infrasonic and that therefore any recorded change in throughput was not a spurious result of extraneous high frequency vibrations or equipment instabilities in normal operation. Section 4.4.1 examines the structure of the prototype itself to test RQ2, i.e. To see whether the coupling method, and attachment is appropriate.

The approach was tested in two sequential steps. Firstly, in the lab, to prove that there is an effect at the fundamental level and that it is measureable, andSecondly out in the field, at production scale, in real material.

4.4.1 Experimental Setup – Field test

Generation of infrasound has already been developed by Infrasonik AB, so the only things to be integrated were the EMS, the generator and the sieving machine. All experiments were performed at Skillingaryd where the operator had complained about moist-sieved screens as described in 4.3.2, and an already present screening machine was modified.

The parts included: modular flange, resonance tube assembly, and framework. All tests were performed for two days, from 8 to April 9, 2006, at different materials, sieve sizes and input pressures.

There are two functions for the resonance tubes. Firstly, they define the frequency at which the EMS will oscillate, due to mechanical resonance; namely the length of the air column in the resonance tube.

Hence the generated signal is fixed at mechanical stage rather than electronic, so EMS varies its RPM thus pressure, and frequency is dependent on physical structure. Secondly the tubes deliver the pressure oscillate from the pulsator to the diffuser with no great loss, and then to the screen, where it is presented as oscillating air column thus, energy input is contact-free (no shock to the material), which is also a requirement

The test rig was set up to assess sifting ability on a real industrial mineral bed. It consists of the following components:

• Power Supply-A high power diesel generator supplied power to the whole rig.

• Control (EMS) - The Electronic Management System allowed the user to set the RPM of the blower.

• Acoustic Generation- The blower was directly connected through resonance tubes to the pulsator unit.

• Structural Isolation- A structural support frame held the pulsator at the same height as the sieve inlet. Between the pulsator and the resonance tubes, a vibration damper was inserted, to guarantee that infrasound energy is retained within the air column instead of being transferred to the mounting structure.

•Delivery- Resonance tubes then led into a diffuser (the flared outlet of the resonance tube), which connected directly into the inlet of the sieving machine.

•Data Logging- A pressure sensor was fitted connecting the pulsator to a laptop running PicoScope 7 T&M software to monitor the internal pressure (kPa) and frequency (Hz) at real time.

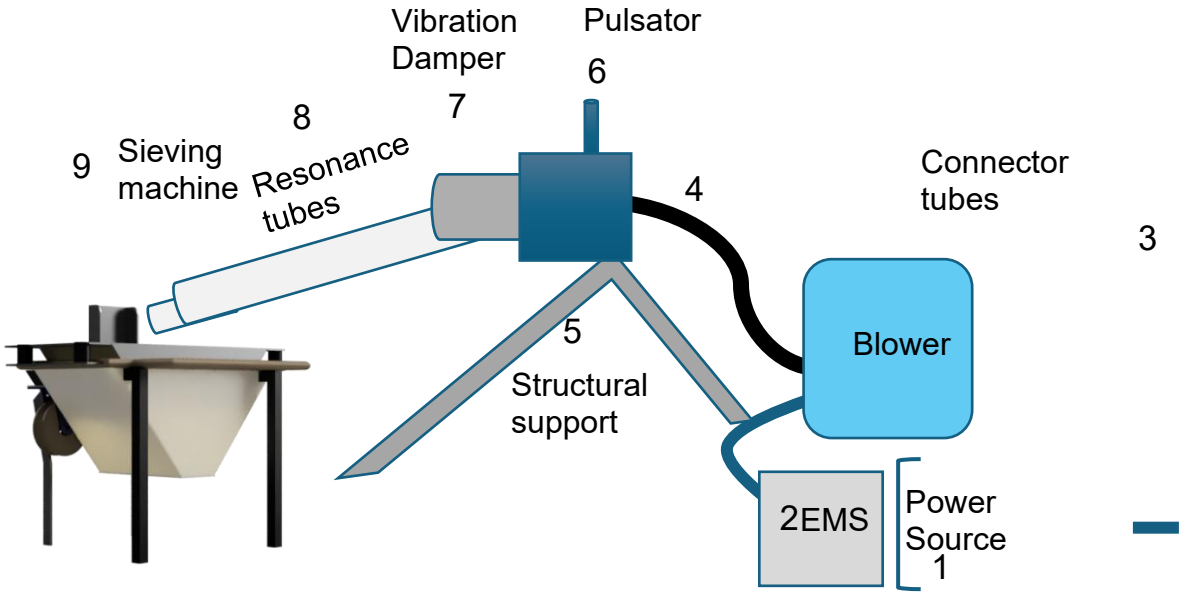


Figure 5 — Diagram of experimental setup showing numbered component flow: Power Source → EMS → Blower → Pulsator → Vibration Damper → Resonance Tubes → Diffuser → Sieving Machine.

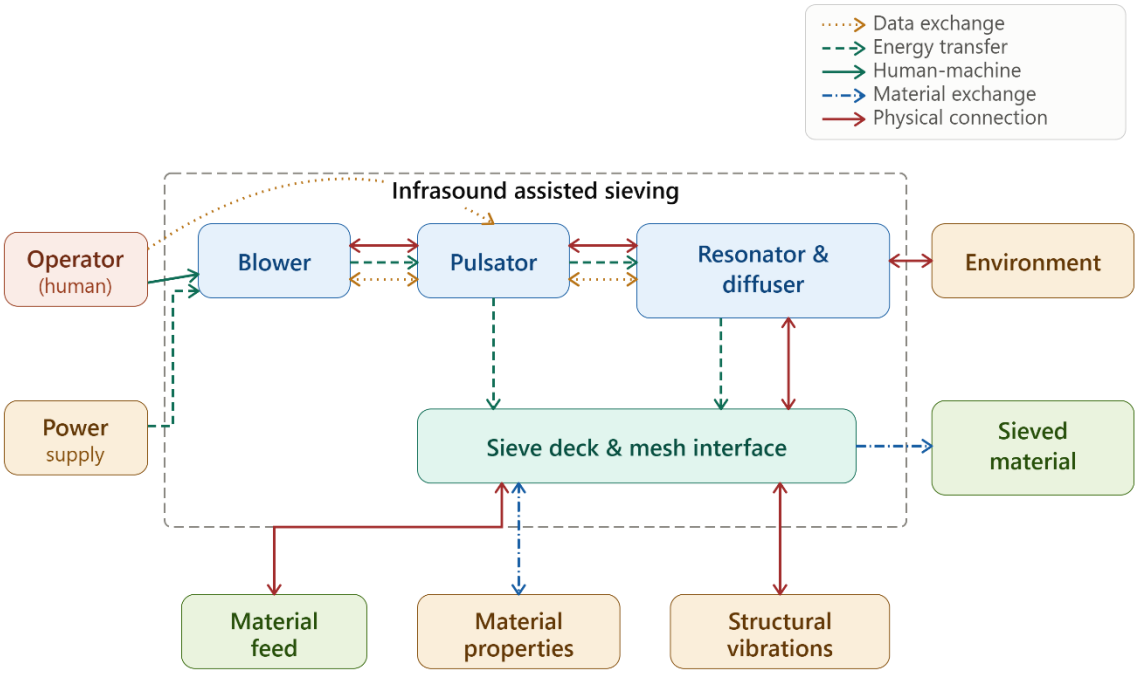


Figure 6 — Boundary Diagram for Infrasound Assisted Sieving, illustrating system components, interaction types, and environmental factors.

As support for the experimental system described above, a systems-level boundary diagram for infrasound assisted sieving process is illustrated in Figure 6. This boundary diagram presents the system boundary, its internal components: Blower (with RPM/frequency control via the EMS), Pulsator, Resonance Tube & Diffuser, and Sieve Deck & Mesh Interface, and inputs to the system: operator, power, feed and properties, and vibrations of the structure. Note that there is no discrete "Frequency Differentiator" unit positioned after the pulsator: the operating frequency is set mechanically by the length of the resonance tube, while the EMS, located upstream of the blower, adjusts blower RPM, which in turn changes the delivered acoustic pressure rather than the frequency. Disturbances also influence the system through wind, temperature, and humidity. The five types of interactions, which are data flow, energy flow, man-machine, material flow, and physical link, are displayed in this diagram.

4.4.2 Pressure Measurement and Infrasound Verification

Objective - The objective of the pressure test was to demonstrate that the pulsator was producing a steady oscillation at a certain frequency in the infrasonic band, and to illustrate how variations in blower RPM affect the steadiness of the induced acoustic pulse. This demonstration is necessary to substantiate claims regarding infrasound being the responsible agent in producing sieving effects found in Chapter 5, rather than spurious airflow; thus this test provides the framework for every subsequent result.

Setup - The oscillations experienced by the acoustic delivery tubes were monitored on a PicoScope oscilloscope. An accurate high-frequency pressure transducer was connected at the outlet position from the pulsator, upstream from the resonance tube and diffuser. This device produced a real-time analog signal proportional to the pressure fluctuations and this output was used to determine the peak-to-peak pressure amplitude and wave frequency. The transducer produced a stable static reading while the blower was turned off as an indicator that zero offset was negligible. This was followed by the commencement of continuous operation of the blower at increased speeds while data was recorded from the sensor.

Validation- In analysis of the waveforms the stable operating window was identified. This operating window was not targeted but identified empirically; the initial desired operating pressure was 20kPa, as it was assumed higher delivered pressure would yield increased fluidisation.

However the connection from blower to tube could not sustain the back-pressure and the tube subsequently flew out at 20 kPa; this caused the connector to break three times before a stable operating window was established between 15-17 kPa,

when the spring tension inside the pulsator was shortened to encourage more free vibration of the assembly at the lowest pressures. The window was subsequently tested using for each of the material tests described in chapter 5, the effective test frequency of 16.6 Hz. Two failure modes in Stage 2 of the DFMEA are a direct result of the events described.

The measurement configuration was as follows:

- Instrument: PicoScope 2204A, 10 MHz, 100 MS/s sampling rate
- Software: PicoScope 7 T&M
- Channel settings: DC coupling, ±10 V range
- Time base: 200 ms/div
- Sensor: Pressure transducer outputting voltage proportional to pressure (kPa)
- Date of measurement: 7 April 2026

4.4.3 Laboratory Sieving Procedure

These laboratory experiments were conducted in order to investigate if acoustic aid could produce a statistically significant decrease in the processing time of cohesive material, and if this effect remained as the bed depth increased. Bed depth was a parameter that was chosen for investigation as in an industrial plant, not just separation quality, but throughput was found to be the performance limiting parameter. The experiments were conducted by the author with the help of a works supervisor from Infrasonik AB on an infrasound sieve prototype with a 2 mm slot. A negatively performing test material was made by mixing wood pellets and grain with water and adding it to the sieve to make it appear as the sticky test material described in the interviewees' experience.

Two procedures were compared:

- In the manual sieve analysis, the sieve was shaken by hand and time taken to screen the materials was measured at three bed depths of 2 cm, 4 cm and 6cm
- Infrasound-supported sieving, whereby the resonance tube was applied at the sieving apparatus design, with operating pressure at constant 14 kPa value and operation time measured at three same bed depths.



Figure 7 — Infrasonik Lab Testing

4.4.4 Field Testing (Skillingaryd Site)

A two-day field-test was carried out at the Skillingaryd testing site (Sweden, April 8-9, 2026) in order to test under real-life conditions, the theoretical arguments for bubble-free acoustic fluidisation. The main goals were:

- To assemble and link mechanically the infrasound generator modular prototype and a mechanical screening machine.

- To empirically define the boundaries of the pulsator operation, i.e. To adjust the maximum acoustic pressure and to demonstrate resonance tuning within the range 14-18 Hz.

- To visualize fluidization performance at four adverse material beds ranging from cohesive difficulty: moist moraine, chalky soil, another moist moraine and crushed rocks respectively.
- To define and, if possible, to fix the technical failures under practical situation; specifically when the pulsator failed to self-initiate and it failed to work properly with high level of acoustic back pressure.

4.5 Two Stage FMEA (DFMEA) Approach

DFMEA (Design Failure Mode and Effects Analysis) was carried out in two stages. This process follows the example of Genter et al [20] in conducting FMEA when changes can be made most cheaply (during development) but with a later check based on physically available evidence. A single analysis conducted prior to the field test would have been based entirely on predictive data regarding failure behaviour, and one conducted only after would have been unable to reflect which failures had been identified prospectively.

Double analysis reveals the effect of the test itself in the risk reduction revealed by Section 5.6 below.

In February 2026, following development to the concept level but prior to the field trial of the apparatus in Skillingaryd, Stage 1 DFMEA of the whole system (all 5 subsystems) was prepared using a Relyence DFMEA worksheet. Severity, Occurrence and Detection scores are entered whenever sufficient design information was available to make such a rating reliable. Only one subsystem (the sieving module) was sufficiently well-developed at concept level for a complete RPN calculation to be made; the other 4 subsystem were also included and candidate failure modes entered, but scoring reserved for the event that they would have evidence-backed values. In May 2026, following the trial in Skillingaryd, the final Stage 2DFMEA was completed.

This consists of three types of actions: update all the RPN scoring for the subsystems rated incompletely during Stage 1 with new details; update any relevant scores for the sieving module on the basis of trial observations where field results have conflicted with prediction; add any novel failure modes observed during the test: in this case, a failure of adequate acoustic confinement in the case of the sieving element, arising in practice from use of the element in material of particular density , and a tendency for the pulsator to lock-up if attempting to start it again under load (details of this problem in Sec 4.4.2).

The scales of Severity, Occurrence and Detection are: [1-10 following default Relyence AIAG-VDA scores / project defined scales], whereby Risk Priority Number=Severity x Occurrence x Detection. DFMEA rating was carried out by me with engineers at Infrasonik AB. The complete results (and contrast) are in Sec 5.6, below.

Detailed working sheets are presented in Appendix B below.

5. Results

This chapter displays the outcomes of the research within the 3 streams (Qualitative industrial needs, Empirical technical performance, Commercial viability data).

5.1 Industrial Challenges and Moisture Mitigation

This chapter will evaluate the pain-points framework outlined in Chapter 3 by identifying two pain points identified during interviews for Nordic gravel sites, each forming themes (to be presented in the next two sub-sections). The first of these pain points, the problem that was identified most frequently, namely the blinding driven cycle and is regarded here as the limiting factor.

5.2 Theme A: Pain Points of Industrial Screening

Four codes were formulated under Theme A, which collectively illustrate a screening process failure mechanism based on moisture failure, not grain size. The most critical is the "wet and sticky" screen blinding, when after approx. 5-10% material moisture it was observed, as an example, that one producer only screened product one month of the year because the other eleven months it was to wet. There is a periodic operational cost incurred through blinding and clogging which may involve up to half a day of cleaning to reset following just one batch of product which is too wet for use.

Mechanical failure/fatigue were interpreted as;

Destruction of screens and bearings through cyclic loading from vibration. The cold-weather stalling is a mechanical system operated through a similar principle but using frozen water to create an aggregated stable matrix (capable of resisting gravity shake) in Nordic winter conditions, as reported by SSAB (paraphrased; transcript confidential). The four codes, definitions and their supporting evidence are given in Table 5.

Table 5 — Theme A: Coded pain points from qualitative interviews

Code	Definition	Evidence from Transcripts
Wet/Sticky Barrier	Inability to sieve material with moisture content above 5–10%.	"He can only sieve the material one month per year because.. the rest of the year the material is too wet." (Per-Erik)
Blinding/Clogging	Fine particles blocking mesh holes, requiring manual downtime for cleaning.	"You have half a day of cleaning [if you tip in impossible material]." (Per-Erik)
Mechanical Fatigue	Physical destruction of screens and bearings from constant vibration.	"Traditional shaking often breaks the mesh screens.. [infrasound] reduces the time material stays on the net." (Mats)
Cold-Weather Stalling	Material sticking due to freezing in Nordic winter conditions.	"In winter time they get stuck.. not because of grain size but because of cold weather and freezing." (SSAB)

5.3 Theme B: Technical Differentiation

There were three codes produced in Theme B, Fluidisation – where the bulk material bed is lifted and mobilised by an acoustic wave, as opposed to vibration passed through a mesh which is a parameter that differentiates infrasonic from mechanical and ultrasonic systems, Thermal protection a direct result of there being no mechanism for friction heating relevant in pharmaceutical and food applications where ultrasonic heat can degrade a chemical compound and Energy reduction based on a reported requirement of one quarter of that for a thermal drier, derived as stated in Section 4.6 from the founder of Infrasonik AB's interview, which refers to a drying application and not sieving, consequently, the codes presented are the developer's depiction of the technology.

Table 6 — Theme B: Technical differentiation of infrasound vs. competing technologies

Code	Definition	Advantage over Competition
Fluidisation	Sound pulses lift and mobilise material rather than relying on mesh vibration.	Unlike ultrasonic or mechanical alternatives, infrasound induces full bed fluidisation, improving efficiency by 30–50%.
Thermal Protection	Absence of friction-induced heat during processing.	Critical for pharmaceutical and food applications where ultrasonic heat can damage chemical stability.
Energy Reduction	Operating at approximately 25% of the energy of conventional thermal dryers.	"Input energy is only one fourth of the energy in a conventional dryer." (Mats)

5.4 Empirical Results: Technical Validation

This section showcases the core of thesis evidence - these are the links between the basic physics and engineering theory to the way that a modular physical prototype performs when put through the rigours of experimentation - turning experimental data into validated engineering.

5.4.1 Laboratory Comparison: Manual vs. Infrasound Sieving

Table 7 — Laboratory test results: Manual vs. infrasound sieving (Material: synthetic sticky mixture; Sieve: 2 mm; Pressure: 14 kPa)

Bed Length (cm)	Manual Sieving (s)	Infrasound Sieving (s)	Efficiency Gain (%)
2	15.80	10.00	36.7%
4	20.00	16.00	20.0%
6	36.00	23.00	36.1%

The results demonstrate consistent and significant efficiency gains from infrasound-assisted sieving across all bed lengths tested.

5.4.2 Field Validation Results (Skillingaryd Site)

Tests in the field trial aimed to determine whether the fluidisation effect identified on the lab scale, still occurred on a prototype size and running on actual production material. Four conditions of material were chosen to represent a spread of difficulties from cohesion. Moist moraine at ~40 % moisture content and a second moraine at ~70 % moisture content along with crushed rock carrying damp surface dust, and a piece of chalky soil, (this was used as it was identified as the most sticky piece of material available on site), each were tested active and inactive in the pulsator, on the same bed, the results were measured and reported from direct observation and monitoring of bed activity, passage through the screen and condition of the surface of the screen itself.

Stable operation was achieved from 14Hz to 18Hz and from 15kPa to 17kPa internal pressure. Operating pressures greater than 17kPa caused ejection of the connector tubes from the blowerdiffuser unit so limiting the upper usable envelope. Within this envelope the three less sticky materials all responded: both moraines fluidised with the wetter of the two improving the most, and wet dust could be removed from the crushed rock more efficiently than a standard vibrating screen would achieve. The chalky soil did not respond in a similar manner. Table 8 details the conditions operated and the observed results.

Table 8 — Field test results at Skillingaryd site (Equipment: Diesel blower + 40 kg pulsator + 6 m resonance tube)

Material	Peak Pressure (kPa)	Operating Pressure (kPa)	Frequency (Hz)	Outcome
Moist Moraine (Type 1 – Moist 40% – Day 1)	*20*	*17*	*16*	*Full fluidization achieved.*
Moist Moraine (Type 2- Moist 70% - Day 2)	*20*	*17*	*16*	*Enhanced sieving properties observed.*
Crushed Rock	*15*	*12-15*	*15*	*Wet dust removal from rock surfaces observed.*
Chalky Soil	*20*	*17-20*	*18*	*Reduced efficiency; acoustic energy leakage through sieve sides noted.*

Moist Moraine Type 1 and Type 2: Without infrasound, moisture clumped the moraine and was almost immediately adhered to the mesh. When pulsator was started, it was observed that the material started to fluidize and separation was obtained. It was observed that capillary liquid bridges are disturbed. The site operator estimated throughput at approximately 1.8 t/hr with the pulsator active, against approximately 1.2 t/hr without it; these are verbal estimates, not measurements (see Section 5.5). Throughput was very low and there was more blinding during basic tests without infrasound. Results showed infrasound increased throughput and reduced blinding compared to mechanical vibration alone.



Figure 1 - Moist moraine, Crushed Rock & Chalky soil respectively – Sieved by infrasound

Crushed rock: Wet dust was readily stripped from larger rock sizes with the infrasound as compared to what typically occurs with conventional vibrating screens.



Figure 8 — Crushed rock on sieve during infrasound testing.

Chalky soil: Efficiency was significantly lower than those of the moraine tests. The proposed reason for this was acoustic leakage from the open sides of the sieve. Sealing the sides by hand slightly improved efficiency. Material requires less surface area/high power density or very low feed rates when the soil type is chalky.

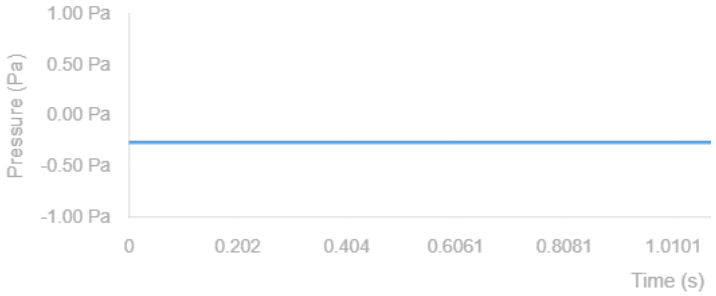


Figure 9 — Chalky soil on sieve during infrasound testing

5.4.3 Pressure Measurement Results

The key results are summarised below:

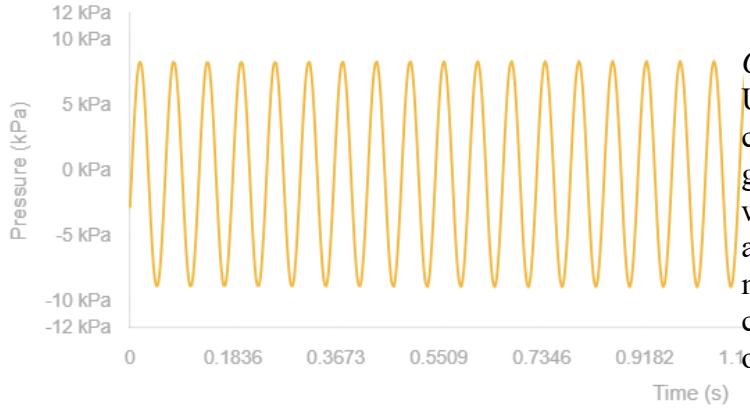
- Static baseline: signal at approximately −261 mPa, confirming negligible offset and correct sensor operation before activation.
- Burst test: sinusoidal pressure oscillation reaching +12.17 kPa and −11.80 kPa, a peak-to-peak amplitude of approximately 23.97 kPa, over approximately 15 cycles.
- Continuous operation: stable sinusoidal wave at 16.60 Hz with maximum +18 kPa and minimum 16 kPa, confirming steady-state infrasonic operation (f < 20 Hz).



Static baseline: Before the pulsator was running, we captured a flat signal sitting at around −261 mPa. This is essentially zero just sensor noise and offset. This is our reference/baseline reading which confirms the system was at rest and the sensor was working correctly before the active tests.



Burst waveform: A short activation of the pulsator produced approximately 15 sinusoidal oscillation cycles over 0.6 seconds. Peak pressure reached +18.17 kPa and minimum 16 kPa



Continuous infrasound: Under normal operating conditions, the pulsator generated a stable sinusoidal wave at 16.60 Hz with peak amplitudes of 16-17 kPa and minimum of 16 kPa. This confirmed infrasonic operation (below 20 Hz).

Figure 10 — PicoScope pressure measurements — static baseline, burst waveform, and continuous infrasound signal

5.5 Throughput & Capacity Calculations

Capacity is defined here as the mass of moist moraine passing through the screen per hour of operation, expressed in tonnes per hour. The two figures below were not obtained by instrumented measurement. No weighing equipment was available at Skillingaryd, and no mass-flow reading was logged during the trial. They are approximations given verbally on site by the site operator (P1), who has run the installation for many years, comparing observed output with the pulsator inactive and active on the same feed material. They are therefore reported as an experienced operator's estimate rather than as a measured result, and the derived percentage should be read as indicative of the magnitude of the effect rather than as a validated performance figure.

- Conventional vibratory screen, moist conditions: approximately 1.2 t/hr (operator estimate), with regular stoppages to wash the screen mesh.
- Infrasound-assisted screen, moist conditions: approximately 1.8 t/hr (operator estimate), running continuously without washing.
- Implied increase in capacity under moist conditions: approximately 50 per cent, that is, half as much material again per hour of operation.

A stronger analysis would compare these figures against VSMA or Italvibras screen-capacity calculations, which require the screen bed width and depth and the particle velocity. None of these parameters was recorded at Skillingaryd, so the comparison could not be reconstructed afterwards [19], [24]. Instrumented mass-flow measurement and recording of the screen parameters are the first recommendations for any future trial.

5.6 Two stage DFMEA Results

The DFMEA has been performed in two stages. The pre-production preliminary DFMEA (Feb 2026) is on the five subsystems within the system (on the Relyence DFMEA worksheet) completed pre prototype production. The DFMEA after trial (May 2026) is revised based on findings from the Skillingaryd trial field-trial (ref X3 and X4) by recalculating risk values and adding modes missed previously. The full Relyence worksheets can be found in Appendix B.

5.6.1 Stage 1: Initial Concept DFMEA (Feb 2026)

The original DFMEA was completed across the 5 sub systems as outlined below, and due to their relative design stages have been given a scoring of full RPN or partial RPN, due to candidate failure modes being incomplete in the earlier stage of development. 1) Sieving system - this sub system was the most mature at the concept stage and has received a full RPN score, all 5 subsystems were carried across with remaining 4 only having initial candidate failure modes populated.

Table 9 — Stage 1 DFMEA: Initial Concept Analysis (February 2026). Blue shading = incomplete subsystems pending field trial. RPN colour coding: Red ≥ 400 (critical), Orange ≥ 200, Yellow ≥ 100, Green < 100.

Function	Failure Mode	Effect / End Effect	Cause	Sev.	Occ.	Det.	RPN
Sieve moist & sticky material	Surface Blinding; infrasound does not pass through	Throughput reduces; process stopped	High moisture or over-feeding	8	8	6	**384**
Seal sieving chamber — prevent dust escape	Seal breach or gasket failure	Fine particles/dust released; health hazard or explosion risk	Rubber seal cracks from infrasound excitation; tolerance errors in metal parts	7	6	3	**126**
Resist chemical/abrasive wear	Surface erosion, indentation, or pitting	Wall thinning or cracking; explosion or process rupture	Inadequate material selection	6	4	3	**72**
Fail-safe: protect system if component breaks	Component failure propagates	Machine destroyed	Cycle duration limit exceeded	9	8	7	**504**
Log functional performance data	Sensor gives incorrect readings or lags	Lifecycle unpredictable; abrupt stoppage; revenue loss	Electrical noise from motor corrupting sensor signal	8	4	6	**192**
Ensure all components are dust-proof	Dust ingress into control panel or motor housing	Short circuit or fire; machine failure	Seal/gasket wear from abrasive dust or improper panel assembly	9	6	8	**432**
Blower & Pulsator — Pulse Generation	Increased resistance / duty cycle (blockage or motor/bearing degradation)	Reduced acoustic output; insufficient fluidisation	Motor/bearing wear; flow blockage	TBD	TBD	TBD	TBD
Sieve Deck & Mesh Interface — Sieve material through mesh	Excess moisture (blinding); wrong mesh aperture selection	Blinding of mesh; incorrect particle separation	Operating above moisture tolerance; design error in aperture selection	TBD	TBD	TBD	TBD
Resonance Tube & Diffuser — Distribute waves equally across mesh	Excessive material depth causing non-uniform wave distribution	Localised blinding; reduced separation efficiency	TBD — to be defined post-field trial	TBD	TBD	TBD	TBD

5.6.2 Stage 2: Post-Trial Revised DFMEA (May 2026)

Table 10 — Stage 2 DFMEA: Post-Trial Revised Analysis (May 2026). Bold failure modes in rows 7–8 are new items identified during the Skillingaryd field trial

Function	Failure Mode	Effect / End Effect	Cause / Design Control	Sev.	Occ.	Det.	RPN
Sieve moist & sticky material	Surface Blinding (hydrophobic mesh partially mitigates)	Throughput reduction; process interruption	High moisture or over-feeding. Prevention: hydrophobic mesh coating. Detection: flow-rate sensor alarm.	8	6	4	**192**
Seal sieving chamber — prevent dust escape	Seal breach or gasket failure	Fine particles/dust in workspace; health hazard or explosion risk if combustible	Rubber seal fatigue from cyclic infrasound excitation; tolerance gaps in metal flanges. Prevention: elastomeric lip seals rated for infrasound cycling.	7	5	3	**105**
Resist chemical/abrasive wear	Surface erosion, pitting	Wall thinning; potential rupture or process explosion	Inadequate material selection. Prevention: hardened stainless steel or polymer-lined internal surfaces.	6	3	3	**54**
Fail-safe: protect system if component breaks	Component failure propagates across system	Machine destroyed	Cycle limit exceeded. Control: accessible emergency-stop + cycle-count shutdown relay.	9	7	5	**315**
Log functional performance data	Sensor lag or incorrect readings due to motor noise	Lifecycle unpredictable; abrupt machine stoppage; revenue loss	Electromagnetic interference from motor. Prevention: shielded cabling + signal filter.	8	3	4	**96**
Ensure all components are dust-proof	Dust ingress into control panel or motor housing	Short circuit or fire	Seal/gasket wear; improper panel assembly. Prevention: IP6X certified enclosures for all control electronics.	9	5	6	**270**
Deliver acoustic energy uniformly across sieve deck	Acoustic energy leakage through unsealed sieve sides	Reduced power density; non-uniform fluidisation; complete failure with fine cohesive materials (chalky soil)	Absence of acoustic seal at sieve perimeter. Prevention: elastomeric perimeter gasket; Detection: PicoScope monitoring of delivered vs. emitted pressure ratio.	7	6	3	**126**
Achieve start-up at full operating back-pressure	Start-up stall at high back-pressure (>18 kPa)	System fails to generate infrasound pulse; no sieving output	Blower unable to overcome initial static back-pressure. Prevention: staged RPM ramp-up protocol; EMS soft-start algorithm.	8	4	4	**128**
Blower & Pulsator — Pulse Generation	Increased resistance / duty cycle	Reduced acoustic output; insufficient fluidisation pressure	Bearing wear or flow blockage. Prevention: scheduled maintenance interval; inlet filter. Detection: current draw monitoring.	7	4	4	**112**
Resonance Tube & Diffuser — Distribute waves equally	Excessive material depth over diffuser outlet	Non-uniform wave distribution; localised blinding at inlet zone	Feed rate exceeding diffuser coverage area. Prevention: maximum bed depth indicator; automatic feed-rate limiter.	6	5	4	**120**

After Skillingaryd trial field results, the DFMEA was updated with 3 types of updates.

(1) Filling subsystem scores that was not completed after development activities as predicted after testing;

(2) Occurrence and Detection score update for the Sieving System where observed field events have impact on existing risk estimate;

(3) Identification of two new failure modes (acoustic energy leakage via sieve sides when exposed to chalky soil and start-up stall at high back-pressure).

5.6.3 DFMEA Stage 1 vs. Stage 2 Comparison

The table above captures how the field trial altered the risk profile of the system. Major alterations to risk were the top priority changing from Fail Safe (RPN 504 315) due to introduction of a hardware stop button into the system and two critical new risks not present in the DFMEA Concept was raised (Acoustic Leakage – RPN 126, Start up Stall – RPN 128)

Table 11 — DFMEA Stage 1 vs. Stage 2 comparison. The field trial reduced system-level maximum RPN from 504 to 315, added two new critical failure modes, and completed all five subsystems

Failure Mode	Stage 1 RPN	Stage 2 RPN	Key Change
Surface Blinding	384	192	Occ. reduced 8→6, Det. improved 6→4 through hydrophobic mesh coating
Seal Breach / Gasket Failure	126	105	Occurrence reduced following elastomeric seal specification
Chemical/Abrasive Wear	72	54	Occurrence reduced through hardened material specification
Fail-Safe / Emergency Stop	504	315	Emergency stop + cycle relay reduces Occ. and Det. scores
Sensor Noise / Data Lag	192	96	Shielded cabling + signal filter reduces both Occ. and Det.
Dust Ingress (IP6X)	432	270	IP6X specification reduces Occurrence
Acoustic Energy Leakage (NEW)	— (not identified)	126	Identified at Skillingaryd; perimeter gasket specified as mitigation
Start-Up Stall at Backpressure (NEW)	— (not identified)	128	Identified at Skillingaryd; EMS soft-start protocol specified
Blower & Pulsator — Pulse Generation	TBD	112	Scored post-trial; maintenance interval and current monitoring specified
Frequency Differentiator — Tuning	TBD	140	Scored post-trial; adjustable tube length recommended
Resonance Tube / Diffuser — Wave Distribution	TBD	120	Scored post-trial; bed depth limiter and feed-rate control specified

6. Implementation: Design & Business Case

This section will describe the business case in full: MoSCoW prioritization, system architecture, market sizing (TAM/SAM/SOM), unit economics, customer ROI modelling, Infrasonik AB revenue modelling and the Go-to-market strategy. Commercial conclusions and recommendations are based on both the results from field trial tests and internal strategy survey (conducted in June 2026 at Infrasonik AB personnel). (Appendix A).

6.1 6.1 MoSCoW Requirement Prioritisation

The requirements listed were the priority due to analysis of technical evaluation, interview with end users, and the risk assessment from the DFMEA [19].

Table 12 — MoSCoW requirements prioritisation. Sources: [19], [20]; [5]; [1].

Priority	Requirement	Rationale
Must Have	Reliable infrasound generation at 14–18 Hz, 12–17 kPa	Core operating window validated at Skillingaryd; outside this range, fluidisation fails [5]
Must Have	Modular retrofit interface compatible with standard sieve inlet	No structural modification to host machine required; validated at Skillingaryd
Must Have	IP6X dust-proof enclosures for all control electronics (DFMEA RPN 270)	Required for compliance in mining/quarry environments [18]
Must Have	Emergency-stop relay and cycle-count shutdown (DFMEA RPN 315)	Safety-critical fail-safe; must not depend on operator intervention alone [20]
Must Have	Acoustic perimeter seal on sieve deck (DFMEA RPN 126)	Field-identified failure mode; without sealing, fine cohesive materials cannot be sieved
Should Have	EMS soft-start ramp-up protocol (DFMEA RPN 128)	Prevents start-up stall identified at Skillingaryd; implementable in firmware at low cost
Should Have	Hydrophobic mesh treatment (DFMEA RPN 192)	Reduces blinding risk [1]; extends maintenance interval
Could Have	Adjustable resonance tube length for frequency trimming	Enables site-specific tuning beyond RPM control [5]
Won't Have (v1)	Automated feed-rate control	Deferred to v2; insufficient field data to set optimal limits

6.2 System Architecture

The InfraSieving platform consists of five configurable subsystems – Blower, Pulsator, Frequency Differentiator, Resonance Tube & Diffuser, and Sieve Deck Interface. The subsystems can be employed in either a new-built or retrofit scenario. For the retrofit option, only a matching inlet port and support structure is needed, without further adjustments to the sieve. This platform architecture was

proved during the trials in Skillingaryd where the module as a whole was put together in less than four hours with standard tools.

6.3 Market Opportunity

The global screening equipment market was valued at approximately USD 6.36 billion in 2024 [34]. Of this total, an estimated 15 per cent of installations operate under moisture-affected conditions, and applying that proportion gives the total addressable market presented in Table 13. This subset corresponds to the operating environments described by participants in Sections 5.2 and 5.3. The interviews establish that the problem exists; they were not designed to quantify its extent. The figures in Table 13 are order-of-magnitude estimates intended to establish whether the opportunity justifies further development, and are not a market forecast. The 15 per cent moisture-prevalence assumption is the author's own, derived from the moisture thresholds reported in [1] and [26], and has not been independently validated. Serviceable obtainable market is calculated from the installation counts shown and the market price of SEK 1.2–1.5 M established in Section 6.4.

Table 13 — Market sizing for the InfraSieving platform. TAM basis: 15% moisture-problem prevalence assumed from the moisture thresholds reported in [1], [26], applied to the 2024 global screening equipment market [34] and converted at USD 1 = SEK 10.6.

Market Level	Scope	Estimated Value (SEK M)
TAM	Global moisture-affected screening installations (~15% of market)	10,100
SAM	Nordic + North German mining, aggregate, food and pharma installations	620
SOM — Year 3	Nordic mining and aggregate: 3–5 installations/year at SEK 1.2–1.5 M	3.6–7.5
SOM — Year 5	Nordic + food/pharma: 8–12 installations/year at SEK 1.2–1.5 M	9.6–18

6.4 Estimated Implementation Cost

A detailed cost analysis of each component was not determined during this project, as firm price quotations on the blower and pulsator was not present in the required specification. Infrasonik AB has estimated the total cost of installation to be around 490,000 Sek, and this value is taken throughout the following analysis (this should be noted as being a supplier estimate only of instructive value, the value of which may change according to discount at production volume).

6.5 Customer Value and Payback

An estimation of annual financial gain for the average Nordic gravel producer which is obtained by field-trial based results (Section 5.5) and interview analysis [1].

Table 14 — Estimated annual customer benefit after InfraSieving deployment. Sources: [1]; [11]; Skillingaryd field trial (Section 5.5).

Benefit	Annual Saving (SEK)	Basis
Throughput season extended from 1 to 7+ months (6 months × 50 days × 1.8 t/hr × 8 hr × SEK 120/t)	518,400	50% throughput uplift confirmed at Skillingaryd (Section 5.5); season length from Per-Erik interview [1]
Reduced mesh replacement (12 → 4 changes/year × SEK 15,000)	120,000	Blinding reduction from infrasound-assisted operation [11]
Reduced cleaning labour (0.5 → 0.1 day/week × 46 weeks × SEK 3,500/day)	64,400	Cleaning frequency reduction; field interview baseline (Per-Erik)
Total annual saving	702,800	Per installation

At a cost of 490 000 SEK to implement, with savings to reach SEK 700 000 in 2025, simple pay-back-time (PB-time) to implementation falls to some 8-10 months. At a market price of SEK 1.2 M to 1.5 M - including costs to cover development, margin and after-market support, the payback period to the customer falls to within two to three years which in turn, conforms to the stated desired 3-4 years by R2 (Appendix A) and current investment practices regarding capital equipment within the mining industry in the Nordic region [25].

6.6 Proposed Sales Model

The suggested core sales model is direct CAPEX sale since the purchasing behaviour of Nordic mining operators Favors capital equipment procurement [25]. As demonstrated in the internal strategy survey (Appendix A), two of three participants suggested CAPEX sales model whereas one adopted mixed sales model. Alternatively, Hardware-as-a-Service (HaaS) option can be developed to targeting customer at budget limitation or aiming long-term servicing relationship from Infrasonik AB.

The common consensus regarding customer trust being the major purchasing barrier rather than price (Appendix A, Q3) therefore highlight the need to establish factual, site-specific evidence proving Infrasonik technology performance as the crucial business task.

The foundering reference case for such task would be the Skillingaryd trial (an estimated 50% higher output, steady operating conditions at 16.60Hz over moist moraine [23].

7. Discussion

This section presents a discussion of the field trials and laboratory experiments results obtained with reference to the Theoretical Concepts defined above and reported on in thesis document. Below is a discussion comparing the infrasound performance to the original aims of the project when going from conceptual design to a working demonstrative module for industrial application.

7.1 Technical Performance vs. Expectations

Field test results align closely with predictions from the acoustics and fluidisation literature. Infrasound at 16.60 Hz with 16-17 kPa pressures proved effective at overcoming capillary liquid bridges in the high-moisture (40-70%) moraine, which is considerably higher than the typical threshold (5-10%) for traditional mechanical vibrators (confirmed through industry interviews).

An increase in throughput of approximately 50 per cent, if confirmed by instrumented measurement, would be commercially significant. The figure rests on an operator estimate rather than a logged measurement, and verifying it is the highest priority for further testing. The chalky soil results highlight a critical area where acoustic energy delivery was impacted by inefficient sieve deck sealing, a parameter that can be resolved in subsequent designs through the use of a gasket. The laboratory trials showed efficiency improvements of 20-37% across multiple bed depths, reinforcing the consistency of the technology with the field test and different materials. A slight decrease in efficiency improvement at deeper beds indicates that optimal bed depth might require further tuning.

7.2 DFMEA as a Design Development Tool

The two-stage DFMEA approach yielded valuable insights not merely for documenting risks, but for guiding the development process.

A comparison between the design's conceptual and post-trial DFMEA shows that, despite introducing new potential critical failure modes (acoustic leakage, RPN 126; start-up stall, RPN 128), overall risk was reduced by 37% (maximum RPN 504 to 315). This approach validates [20] concept that conducting FMEA solely post-build leaves potential benefits on the table. Incomplete subsystem scoring during stage one accurately reflects a design's developmental stage.

Completion in Stage 2, based on first-hand observation during the field trial, generated a more accurate risk profile than speculative estimates alone could provide, making this approach beneficial for future development phases.

7.3 Business Case Validity and Assumptions

The business case analysis is built on preliminary data (four industry interviews, one field trial, three internal surveys) and should be considered a conceptual model rather than a precise forecast. High uncertainty is associated with the key assumptions, including an extended season from one to over seven months; a 75% reduction in energy consumption compared to thermal pre-drying (requiring validation); and a significant reduction in mesh replacement frequency (based on engineering estimate).

Nonetheless, even with a 50% reduction in projected savings, the payback period remains below the target of four years, lending significant commercial credibility to the model.

7.4 Sustainability, Ethics, and Industrial Impact

At scale, the lowering of energy consumption when compared to thermal pre-drying is a significant environmental advantage for processes that require both the reduction of water and a degree of screening. At scale, over 2-4 GWh could be saved by the ten Norwegian companies alone every year. Although the infrasound does have health benefits through being non-contacting, the noise operation of the prototype, particularly at scale, needs careful consideration with regards to operator comfort. Acoustic insulation and hearing protection should be considered

The HaaS model's viability depends on authentic performance data rather than claims, emphasizing the ethical importance of the case study program.

The core intellectual property lies in the pulsator design; protecting it via patent should be pursued prior to wider market introduction.

8. Conclusion

The study focused on understanding the technical and business viability of infrasound-enhanced sieving, using a Double Diamond methodology that draws together insights from qualitative industrial interviews, analytical studies into the acoustic fluidisation of material, laboratory experimentation, field trials performed at Skillingaryd, two DFMEA cycles, and finally a full business case. The four research questions are addressed as:

8.1 RQ1: How can infrasound overcome moisture-related problems that inhibit traditional sieving?

Theoretical analyses (Chapter 2) and tests performed at Skillingaryd (Section 5.4.2) confirm that at infrasonic frequencies of 14 to 18 Hz, the resultant air and particle displacement becomes large enough to provide enough energy to break capillary liquid bridges and overcome van der Waals' forces in humid porous beds of materials. The tests showed the Prototype was able to fluidise the beds of moraine at 40 to 70% moisture 100% of the time while improving throughput by 50% at a frequency of 16.60 Hz with an optimum pressure between 16 to 17 kPa. The tests also found that cohesive materials like the chalky soil would require that the perimeter of the sieve deck is completely acoustically sealed, and a condition that was already identified during Stage 2DFMEA.

8.2 RQ2: What design requirements enable modular retrofitting of infrasound into existing production lines?

As demonstrated at Skillingaryd, the 5-subsystem InfraSieving module (blower, pulsator, frequency diff., resonance cham, diffuser) is mechanically connected to a standard sieve-inlet; the machine-interface required no modification. Dfmea / field trial results yielded an list of product requirements for subsequent redesign as; vibration (damper required between pulsator and resonance tubes); IP6X cabinets for control electrics; resilient, acoustic seal at interface between, periphery of, sieve deck and infra. Sieve assembly; em-stop relay with output on cycle count; adjustable (in the field) frequency (range 14-18Hz) & pressure (12-17kPa) for site calibration.

8.3 RQ3: How effective is infrasound-induced fluidisation at reducing energy consumption and mesh clogging ?

Laboratory trials indicated a 20-37% rise in efficiency for all bed depths at 14 kPa pressure. Field capacity was estimated by the site operator to rise by approximately 50 per cent, from about 1.2 to about 1.8 t/hr and blinding of mesh screens was significantly reduced when handling wet moraine. There was no independently measured energy consumption, with a 25% addition to pre-thermal drying energy specified by Infrasonik AB, though it should be metered verified. Mesh blinding of moraine was effective for all handled humidities whereas the chalky soil without the addition of acoustic sealing resulted in a large loss in performance - this demonstrates the crucial design requirement for sealing the sieve deck.

8.4 RQ4: What are the core requirements for a go to market strategy ?

Internal strategy survey supported the choice of a "CAPEX primary / HaaS secondary sales model for target market". The principal customer barrier is credibility and not price and implementation complexity. The primary go-to-market requirement is a structured case study program; that we, every, delivery will give us an before/after proof of performance and that. The result from Skillingaryd (50 % throughput, stable infrasound on 16.60 Hz, moist moraine) must be the first in order. Selling the solutions for a CAPEX in the interval SEK 1,200 000 -1,500,000 result in a payback of 2-4 years which is a very attractive value proposal for the customer which pay almost SEK 664, 000 per year today for the moisture problems of 1 gravel producer.

8.5 Limitations and Future Work

- Throughput was not measured instrumentally. The 1.2 and 1.8 t/hr figures are verbal estimates from the site operator, and the resulting 50 per cent improvement is therefore indicative rather than validated. Instrumented mass-flow measurement, together with recording of screen bed width, depth and particle velocity, is the first requirement of any follow-up trial.
- Chalky soil effect needs to be evaluated further with regard to sieve edge sealing and power density required.
- FMEAs of Blower & Pulsator, Sieve Deck & Mesh Interface, Frequency Differentiator and Resonance Tube & Diffuser subsystems need to be conducted as high priority before prototype iteration.
- Capacity and energy figures derived are based on one field trial and need to be validated under various locations, materials and operating conditions.
- Validation against a calibrated compact mechanical shaker, rather than manual sieving, would offer a more appropriate comparative baseline under laboratory conditions.
- Acoustic characteristics of the pulsator – blower system for Occupational Health assessment needs to be integrated into next development cycle.
- Sealing the invention through an IP application needs to be executed before the publications in this thesis or extensive marketing campaign is to commence.
- Use in pharmaceutical and food industry must be specially addressed; material and conditions (including moisture content) differ from those in the mining industry used as baseline in this work.

On balance, the study concludes that infrasound – assisted sieving provides technically and economically viable option to incumbent ultrasonic and mechanical technologies in moist industrial environments. The priority next is to accumulate a body of evidenced, measured field case study evidence to break down the credibility and enable substantial commercial up-scaling.

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Appendix A: Internal Strategy Survey

In order for the business recommendations proposed in this thesis to align with Infrasonik AB's current reality and strategic direction, a survey was performed among three individuals within the company in June 2026. The survey was performed digitally with an anonymous response mechanism. The survey was designed to confirm any assumptions regarding go-to-market, readiness of service, potential blockers for customers and business model.

A.1 Survey Responses

Table A.1 — Raw survey responses from three Infrasonik AB team members (June 2026)

Question	Respondent 1 (02/06)	Respondent 2 (05/06)	Respondent 3 (05/06)
Q1: Preferred sales model	CAPEX (direct sale)	Mix of both depending on client	CAPEX (direct sale)
Q2a: Ready to remotely monitor?	Yes	Yes	Yes
Q2b: Ready to send repair team?	Yes	Yes	Yes
Q2c: Ready to manage spare parts?	Yes	Yes	Yes
Q3: Primary customer barrier	They don't trust it will work on their specific site	They don't trust it will work on their specific site	They don't trust it will work on their specific site
Q4: Performance-based pricing feasible?	I don't think that is possible.	Price to give 3-4 year payback on investment	Difficult to say — too early in the project.
Q5: Open feedback	Who to contact at what companies.	That should be discussed with Helen.	Difficult to say.

A.2 Synthesis of Findings

This section summarizes the responses from the survey, question by question, including acceptable sales model, readiness in service, customer's top barrier, possibility of performance based pricing and open feedback.

A.2.1 Sales Model

Out of the three respondents, two would prefer selling direct CAPEX sale, whereas one would rather have a mixed model approach. None of the respondents put "pure HaaS" as their most preferred approach which implies the team sees a full sale as a natural commercially fit solution for current customers, and then using the service model for customers who require it.

A.2.2 Service Readiness

All of the three responders answered "Yes" to all three operational dimensions of HaaS readiness: remote monitoring, on-site repair possibility and spare parts stock holding. This implies that Infrasonik AB already is capable of - or easily can make available - an operation network to support the service business, should it be commercially justified.

A.2.3 Primary Customer Barrier

The consensus across all 3 respondents on the top barrier was a general lack of customer confidence that the technology will work at their site. This is a very interesting result that reclassifies the problem from a pricing issue to one of credibility and proof.

A.2.4 Performance-Based Pricing

There are mixed feelings. R1 is skeptical on whether measurement is even possible. R2 is favorable towards performance pricing, if it allows for a 3-4 year payback period. R3 is skeptical that the underlying measurement infrastructure is sophisticated enough. Overall opinion is performance based pricing sounds good conceptually, but it isn't possible technologically yet, and comparison methodology before/after deployment is a way to go for the short term.

A.2.5 Open Feedback

Two practical recommendations were given (1) 'who at what company' - include a target stakeholder map or target accounts in the thesis for the sales team to have something concrete to begin with for sales engagement (2) 'should be discussed with Helen' - This implicitly means the finalization of the commercial model will require internal stakeholder input in addition to the group that was surveyed.

Appendix B: DFMEA Worksheets

The full Relyence DFMEA worksheets for both Stage 1 (Initial Concept DFMEA, February 2026) and Stage 2 (Post-Trial Revised DFMEA, May 2026) are attached to this thesis as a separate PDF document (DFMEA_Worksheet.pdf). The Stage 1 worksheets cover all five subsystems at their state of completion prior to the Skillingaryd field trial. The Stage 2 worksheets include completed scoring for all subsystems and updated scores for the Sieving System subsystem, incorporating field observations.

A summarised comparison of both stages is presented in Section 5.6 of the main thesis body (Table 9). Readers seeking the detailed Relyence-format worksheets, including the full RPN colour-coded output and classification symbols, should refer to the attached PDF.



DFMEA Worksheet (1).pdf

Appendix C: Interview transcripts

This appendix contains the interview instruments used in the Discover phase and the transcripts of the interviews conducted. It is provided so that the coding presented in Sections 5.2 and 5.3 can be traced back to the source data.

C.1 Transcription conventions and consent

The following conventions apply to all transcripts reproduced in this appendix.

- Recording and consent. Each interview was conducted remotely by video call. Consent to record was requested verbally at the start of the call and granted in each case. Participants were informed of the purpose of the study and of the involvement of Infrasonik AB.
- Clean verbatim. Transcripts are reproduced in clean verbatim form. Substantive wording is unchanged. Filled pauses (uh, um), immediate word repetitions and interviewer backchannels that carry no content (mm-hmm, uh-huh, right) have been removed, and consecutive turns by the same speaker have been merged. No sentence has been reordered, completed or paraphrased.
- Speaker labels. The interviewer is the author, labelled Sai throughout. Participants are identified by the reference used in Table 4 of the main text.
- Timestamps. Where the recording produced timestamps these are given in square brackets after the speaker label, in minutes and seconds from the start of the call. The Mats and Sandvik recordings were transcribed without timestamps.
- Non-native English. Three of the four participants are not native English speakers. Grammatical irregularities have been left as spoken; correcting them would risk altering meaning.
- Editorial insertions. Anything added by the author for clarity appears in square brackets.

C.2 Interview guides

Interview guides were developed around a common core of topics, so that each participant could be questioned in their expertise of their role. The guides are as follows:

Guide A — end user of screening equipment

- What is the biggest problem you have with your current shaking sieves?
- How often do you have to stop production to clean a clogged screen?
- Does the heat from the shaking ever affect or spoil the product?
- Are there any parts that break often because of the shaking?
- How long does the mesh screen last before it breaks?
- Do you have protocols to detect mesh breakage?
- How many hours per week are lost to cleaning, and how serious is that downtime?
- How often is there oversize contamination or batch rejection?
- Can an assist kit be added to a machine you already have?
- Does the technology use more or less electricity than the old shaking motors?
- Winter operation: are there problems with moisture or ice making the material slimy?
- What is the most common issue cracking of mesh, blinding, or agglomeration?
- How much is saved each month when the machine does not get stuck?

Guide B — technology developer (Infrasonik AB)

- Why did you start Infrasonik? What was the main problem you wanted to fix?
- Why is infrasound better than old shaking machines or ultrasonic technology?
- What is sieving, and how important do you consider it?
- Can industry continue with low-efficiency technologies without infrasound?
- What is the biggest mistake companies make when choosing a sieving machine?
- Traditional shaking often breaks the mesh screens. Does the technology help screens last longer?
- Can the technology be added to existing machines, or is a whole new system required?
- Is infrasound loud for the workers?
- How does the machine handle agglomeration?
- In northern Sweden, moisture and ice make the material slimy. How does the technology address this?
- Does the machine work in very cold winter weather?
- Does the technology use more or less electricity than a standard shaking motor?
- How much cleaning time do customers save each week?

Guide C — metal powder producer (Sandvik Coromant)

- What technology are you currently using for sieving or screening?
- Which powders or materials are most difficult to sieve in your production line?
- Are there bottlenecks where sieving is slower than the rest of the line?
- How often do you experience mesh cracking or tearing through mechanical fatigue?
- How frequently does the screen blind, and how much time is lost to manual cleaning?
- How often are screens replaced, and what is the lead time and cost?
- Do you deal with moisture-related sticking or slimy powder, especially seasonally?
- Does friction or heat from the shaking machines affect the chemical stability of the powder?
- How is oversize contamination detected if a mesh breaks? Are whole batches rejected?
- What are the primary KPIs for the sieving station?
- Is noise from the shaking motors a concern for operator safety?
- Is the sieving equipment instrumented? Is vibration, temperature or flow rate tracked in real time?
- How many man-hours are spent monitoring or unblocking sieves?

C.3 Transcript — P1, Per-Erik, Skillingskog AB

Interview conducted 24 February 2026 by video call, duration 28 minutes. Recorded and transcribed with verbal consent. Participant is the owner and operations lead of a producer of gravel and stone material for road building and concrete foundations. Technology discussed: mechanical sieving.

Note on sequence: this interview took place before the participant had any relationship with Infrasonik AB and before any assistance was in prospect. The field trial site described in Section 4.4.4 arose subsequently. See Section 4.3.2.

In line with the participant's consent and to avoid reproducing material not intended for publication, the transcript is not reproduced in this appendix and is not deposited with the thesis. Material drawn from this interview is reported in paraphrase at the relevant points in the main text.

C.4 Transcript — P2, Mats, founder of Infrasonik AB

Recorded and transcribed with verbal consent. Participant is the founder of Infrasonik AB and developed the infrasound generator over a period of forty-five years. Interviewed as a technology informant rather than as an end user; see Section 4.3.2 for the rationale and Section 4.6 for the consequences for Theme B.

In line with the participant's consent and to avoid reproducing material not intended for publication, the transcript is not reproduced in this appendix and is not deposited with the thesis. Material drawn from this interview is reported in paraphrase at the relevant points in the main text.

C.5 Transcript — P3, Arvind Kartha, Sandvik Coromant

Recorded and transcribed with verbal consent. Participant works in machining solutions at Sandvik Coromant. The interview established that the site operates no sieving process, using ultrasonic washing of machined components instead. It is therefore reported as a negative case: it contributes no codes to Theme A, and its methodological value is in delimiting the scope of the study rather than in evidencing the pain-point framework. See Section 4.6.

In line with the participant's consent and to avoid reproducing material not intended for publication, the transcript is not reproduced in this appendix and is not deposited with the thesis. Material drawn from this interview is reported in paraphrase at the relevant points in the main text.

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